

Digital Marketing

Module III

Mobile Marketing
Marketing Analytics & ROI

Syllabus

3. Mobile Marketing

- Mobile landscapes for Marketing and Monetization,
- Conventional Advertising,
- Millennial Mobile Advertising,
- Versatile Promotions,
- Alternative focusing and promotions on Mobile,
- Push App and Game based promotions,
- Location evolution with mobile

3.2 Marketing Analytics & ROI

- Key marketing engagement & ROI metrics,
- Primer on data science and analytics concepts,
- Web Traffic nuances,
- Multi-channel Analytics,
- Decoding CLV and RFM,
- Deciphering conversion and goal metrics,
- Implement conjoint analysis & decision tree tactics,
- Avoiding common analytical pitfalls.

Digital Marketing & Transformation (BCAE0005)



Lecture -25

Anuj Mangal

anuj.mangal@gla.ac.in

+91-9897534383

3.1 Mobile Marketing

Interesting Facts

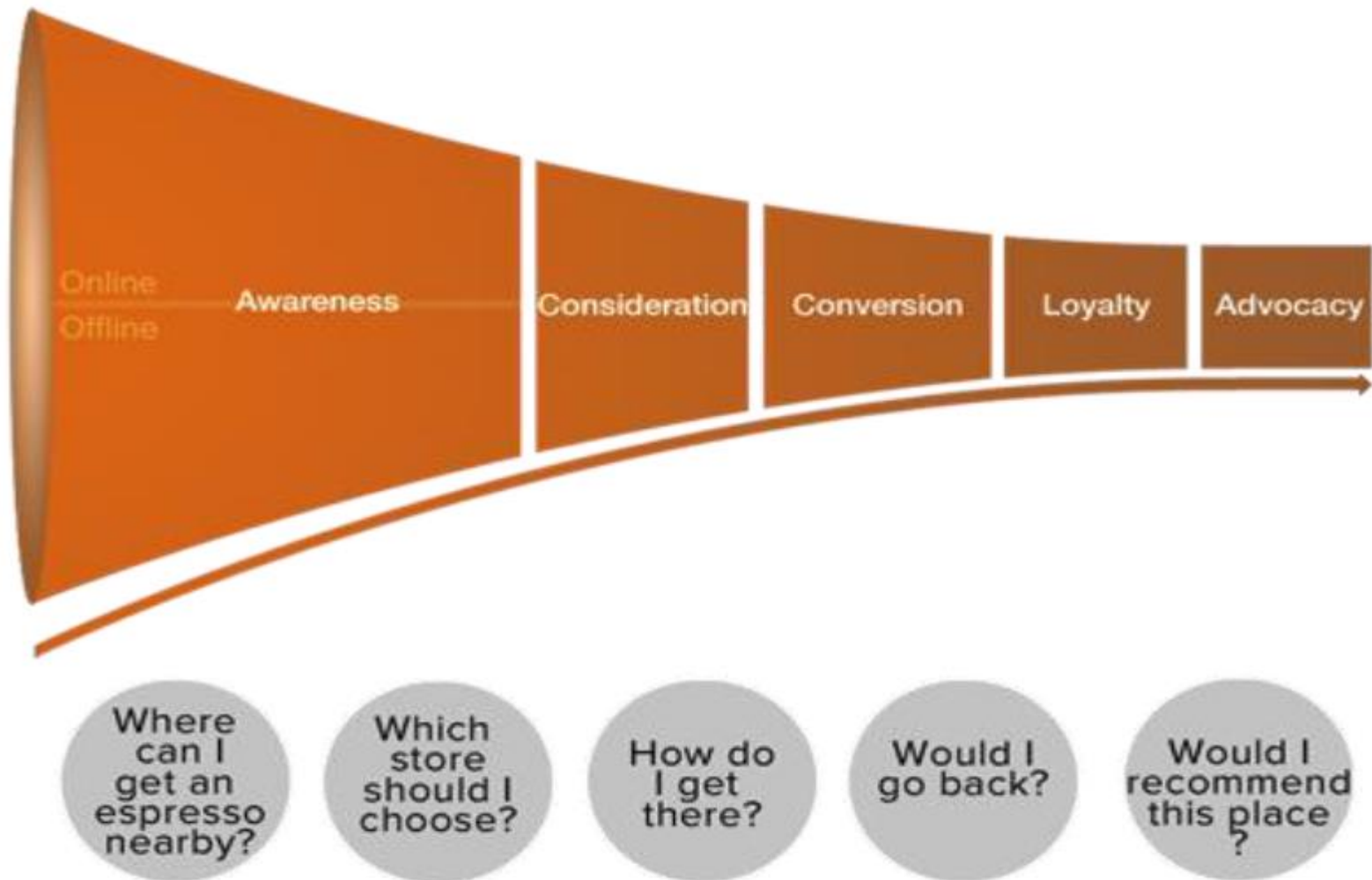
- By 2019, mobile advertising is expected to represent 72% of all U.S. digital ad spending
- As of March 2017, 80% of top websites were mobile compatible
- 48% of consumers start any mobile informational journey via an organic search compared to
- 33% of consumers whose journey is attributed to direct traffic (i.e. indicative of deeper engagement).
- Google drives 96% of mobile search traffic, followed by Yahoo at 2% and Bing at 1%.

- Gaming apps represent the largest percent of mobile usage, with a consumption of over 80%
- Web engagement is 70% higher on tablets than desktop & smartphones
- Mobile searches and app usage have increased 200% year over year in 2017 and have already surpassed desktop searches in 2015

3.1.1 Mobile Landscapes for Marketing and Monetization

- **Mobile Marketing:** is a multi-channel, digital marketing strategy aimed at reaching a target audience on their smartphones, tablets, and/or other mobile devices, via websites, email, SMS and MMS, social media, and apps.
- [80% of internet users](#) own a smartphone.

Mobile Purchase Decision Funnel



Mobile Marketing Strategy

3.1.2 Conventional Advertising

- One of the key differences between traditional marketing and mobile marketing is that traditional marketing has historically been used primarily for creating a brand preference and driving demand.
- Mobile, on the other hand, takes it up a notch to facilitate **ease of transactions** in addition to the above factors.
- A mobile consumer has a larger tendency to convert than a non-mobile consumer.

- **SMS and MMS Marketing Is Conventional**
- This can be unearthed from the fact that email open rates on mobile channels are about 60% on average however a text message is read almost 90% of the time.
- Thereby, it is essential to take adequate advantage of the opportunities provided by this channel, to increase audience engagement in SMS campaigns.

- Note that it is neither ethical nor legal to send unwanted messages with SMS marketing.
- SMS and MMS are influential channels for mobile marketing.
- About 4 billion people are able to receive SMS messages, and 90% of those messages are opened within three minutes.
- Consider the following:
 - The open rate of SMS is 95% as compared to 22% for emails.
 - There is an observed lift of 50% in sales conversion when a user is sent a marketing SMS.
 - Text messages can be 8x more effective at engaging customers as compared to other channels.

- SMS/MMS is highly personal as compared to other marketing channels, so thereby it is key to ensure that our messaging is always thoughtful and non-intrusive.
- Best Practices for SMS
 - Text limit should be of 140-160 characters.
 - Avoid using jargons.
 - Make the message valuable.
 - Include sender profile
 - Include a clear call-to-action.

- MMS is an advanced version of SMS marketing, where-in messages can be sent peer-to-peer, a mobile messaging service provider or a website to any mobile phone.
- MMS messages can include text, photos, videos, audios, GIFs etc.

Why should MMS be preferred over SMS?

- Increases campaign opt-ins by 20% over SMS.
- MMS texts have a higher customer engagement with a 15% average CTR (i.e. clickthrough rate).
- It is likely to be shared on social platforms.
- It is advantageous to explore the additional media options that comes with leveraging MMS messaging:
 - Include product-oriented visuals
 - Link the MMS to other marketing channels/campaigns
 - Provide CTA for social sharing

Regulatory Compliance

- Since these messages are considered as automated sends, it is of paramount importance to consider privacy regulations while plan SMS and MMS campaigns.
- Below are the three privacy principles that should oversee the implementation of SMS and MMS marketing:
 1. Adequate notice— customers should be informed that they will be receiving SMS messages from a concrete shortcode based program.
 2. Opt-in consent— getting explicit opt-in confirmation from users before sending any marketing SMS and MMS messages.
 3. Opting out— It should be very clear how someone can opt out of the program.

Digital Marketing & Transformation (BCAE0005)



Lecture -26

Anuj Mangal

anuj.mangal@gla.ac.in

+91-9897534383

Conventional Mobile Advertising Strategy

Step 1 - Create Mobile Buyer Personas:

- Creating and implementing buyer personas are valuable mechanisms to deduce target audience.
- Deciphering a target audience is the key to any marketing strategy.
- They can be construed primarily as fictional depictions of the various customer profiles.
- The key attributes to consider while designing a buyer persona (apart from personal details) are as follows:
 - Background
 - Job description
 - Source of information
 - Goals
 - Challenges
 - Preferred content

- It is essential to collect mobile behavioral data of the audience and try to gather valuable actionable insights and usage patterns.
- Some interesting observations include:
 - Over 60% of all emails are first opened on a mobile device.
 - Around 48% of users start their mobile internet sessions via a search engine.
 - Nearly 95% of adults primarily use their smartphones to access content/information.

Step 2 - Set Goals

- Determining the key performance indicators and success criteria are key for defining any effective marketing strategy.
- Thus, it is very important to consider views of all stakeholders to map these elements.
- The following activities can help us to achieve these objectives:
 - Defining time-lines to ensure everyone is on the same page
 - Describing the reasons and expectations for considering mobile marketing
 - Evaluation of all the previous marketing initiatives and their associated performances to showcase the reaped benefits

Step 3 - Establish KPIs (Key Performance Indicators)

- It is essential to lay out the key performance indicators that are realistic.
- The successful outcomes should depend on measurable campaign criteria.
- The following factors should be considered for defining the KPI landscape:
 - Conversion rate
 - Cost per acquisition
 - Retention rate
 - Churn rate
 - Daily active users (DAU)
 - Monthly active users (MAU)
 - Return on investment

Step 4 - Monitor Performance Metrics

- Performance analytics determine the key success factors of any marketing campaign which stands true for mobile campaigns as well.
- Thereby to ensure success of any campaign, it is essential to measure the associated KPIs against the set goals, understand relevant trends and behaviors and convert them into actionable insights wherever applicable.



Active Users



How are your users
navigating?



How long does a user
stay on your app?



User retention



How much does it cost to
acquire a new customer?



Average revenue
per user (ARPU)



Customer Lifetime
Value (LTV)

Monitor Performance Metrics

3.1.3 Millennial Mobile Advertising

- Millennials are the generation that truly grew up during the tech evolution, stretching back from the days of dial-up internet access to the current smartphone friendly generation.
- Newsweek magazine reported that the Millennial generation was born between 1977 and 1994.
- They are true digital natives and are considered to be a tech-savvy and a hyper-connected group that we all know today.

- As per ComScore, below are some facts which we should consider while strategizing mobile marketing for millennials.
 - Millennials hate push notifications, but have only themselves to blame as they are usually not mindful of basic app permissions.
 - Millennials don't mind shelling out money for apps.
 - Millennials will delete an app if its logo is unappealing.

- A mobile-friendly website is no longer an option—it's a must.
- The rise in mobile traffic coupled with Google's mobile-friendliness ranking factor.
- For search engines, “mobile-friendliness” suggest to the following:
 - Content fits on the screen without side-to-side scrolling or zooming.
 - Optimum page load times
 - Website returns no mobile-specific errors.

- The most important reason to maintain a mobile-friendly site is to create a consistent and engaging user experience.
- Mobile UX always has a dramatic effect on every stage of the buying cycle:
 - 64% of mobile web users discard pages if they don't load within 10 seconds.
 - 35% of executives could not make an intended purchase because the website they visited wasn't mobile-friendly.
 - 90% of the executives use mobile devices to research potential business procurement.

- Key considerations for designing email CTAs for mobile users:
 - Use responsive design—a methodology that automatically formats web page content for optimal viewing on any device
 - Ensure CTA visibility (possibly at the top).
 - Ensure optimal sizing of the CTA button, so they are easily “tap-able.”

Digital Marketing & Transformation (BCAE0005)



Lecture -27

Anuj Mangal

anuj.mangal@gla.ac.in

+91-9897534383

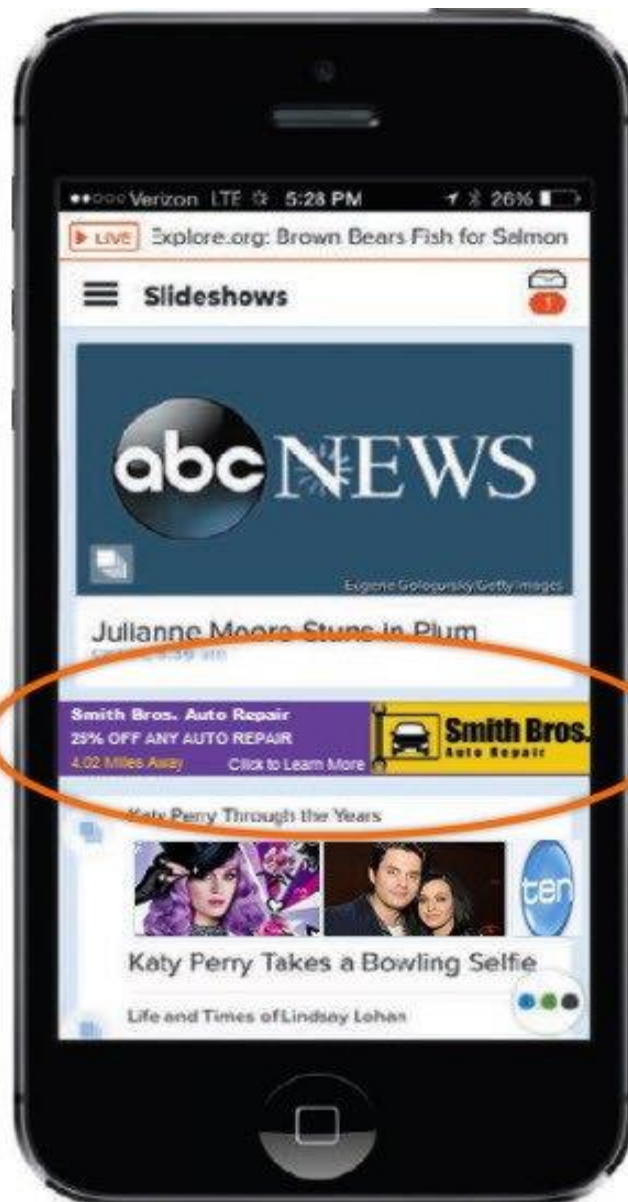
3.1.5 Alternative focusing and promotions on Mobile

1. QR codes:

- QR codes are scanned by users, who are then redirected to a specific webpage, embedded within the QR code.
- They are often aligned with mobile gamification principles and have an element of probabilistic chance to them, this appeals to inherent human psychological desires for attaining reward, self expression, achievement, competition, and status.
- <https://www.qrstuff.com/>

2. Mobile search and image ads

- These are basic Google search ads built for mobile, often featuring extra add-on extensions like click-to-call or maps.
- Image-based ads designed to appear on mobile devices.



3.1.6 Push App and Game based promotions

App-based Marketing

- Mobile apps can help in accomplishing various marketing goals like:
 - maximizing brand awareness for the product,
 - driving engagement,
 - and supporting online sales activities.

- It is also essential to ensure that the app encourages user engagement to build relationships and loyalty, and drives conversion.
- These conversions are driven by two types of messages:
 1. Push notifications.
 2. In-app notifications

Push Notifications

- Push notifications are an effective marketing tool for anyone with a mobile app because they help in engaging regularly with users.
- Push notifications look like SMS text messages and mobile alerts, but they only reach users who have installed the app.
- Each mobile platform has support for push notifications (i.e. iOS, Android, Fire OS, Windows and BlackBerry, with their own services).

- Push notifications provide convenience and value to app users.
- For example, users can receive:
 - Stock updates and news right on their lock screen
 - Utility messages like traffic and weather reports
 - Flight check in, change, and connection information

- For app publishers, push notifications are a way to speak directly to a user.
- These messages usually don't get caught in spam filters or forgotten in an inbox — and furthermore have click-through rates which can be twice as high than an email.
- They can also be used to drive actions, such as:
 - Promoting products or offers to increase sales
 - Improving customer experience
 - Converting unknown app users to known customers
 - Driving users to other marketing channels, such as social networks

In-App Notifications

- In-app notifications are displayed while the user is active within the app itself – also referred to as native app content.
- The in-app messaging is highly appropriate.
- They are rooted in analytics and can be triggered based on user actions.
- Because of its personalized style, in-app messaging is better at delivering content, based on user context and expectations.
- In-app messaging should feel like a natural part of the app not additional marketing.

- Advantage of in-app notifications:
 - Introducing new app features to users.
 - Sending messages to promote engagement with specific content pieces.

Getting listed on app stores

- Sign-up with Google/Apple or release on own website
 - Installation required in APK/IPA format
 - For non-store apps, the allow untrusted sources configuration should be set on device
 - Quality check (Guidelines provided for each of them)
 - UI and usability
 - Functionality (it should work)
 - Size issues

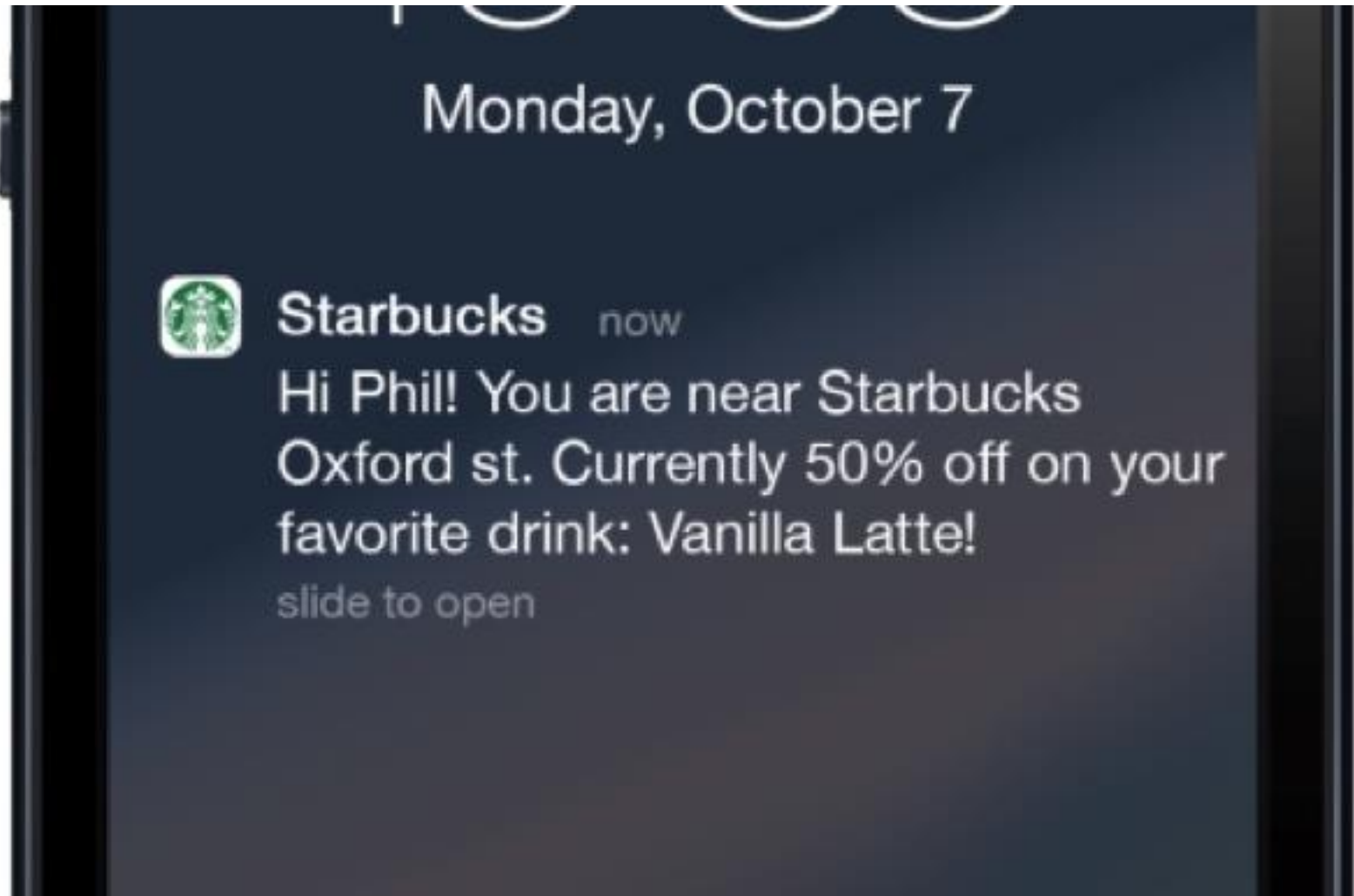
App Store optimization (ASO)

- Optimizing on Search Engines = Search Engine Optimization
- Optimizing to be found on App Store = App Store optimization
 - Keywords are different (update to show latest and changing user search queries)
 - Fetching Customer Reviews
 - Smart looking screenshots
 - Tying back with Google+

App Store Optimization (ASO) Guidelines

- Key points to focus on for ASO (App Store Optimization)
 - Keep the user in mind
 - App name should be unique
 - Description should contain typical usage of the app
 - Focus on relationships and not on transactions
 - Use customer feedback for improving the product

3.1.7 Location Evolution with Mobile



Location-based marketing

- Ads that appear on mobile devices based upon a user's location relative to a specific area or business.
- For example, some advertisers may only want their mobile ads to appear when users are within a one-mile radius of their business establishment.

Best Practices for Mobile Marketing

- Consider making emails, ads and sites for mobile **mostly text based** to ensure **deliverability and readability** of the message.
- *Smaller blocks of text, shorter sentences = more seen, read, and understood in a shorter amount of time*
- Make sure the mobile apps or optimized sites have very **clear call to action prompts and are easy to use**
- The **fewer obstacles in sharing**, the happier the consumers will be
- **Optimize** the campaigns for **frequently misspelled search words**

Digital Marketing & Transformation (BCAE0005)



Lecture -28

Anuj Mangal

anuj.mangal@gla.ac.in

+91-9897534383

3.2 Marketing Analytics and ROI

Interesting Facts

- People on an average nowadays, spend more time online than on TV and other media (newspapers, magazines, etc.) combined.
- Total spending on Internet advertising is predicted to grow 12.9% next year.
- The Internet will become the largest medium for advertising in 2019 (ahead of TV).
- Less than 30% of small businesses use Web site analytics.
- 18% of small businesses admit that they do not tracking anything.
- Users who are retargeted to, are more likely to convert by 70%.

Analytics and ROI – The Connection

- **Marketing analytics** is the practice of
 - measuring,
 - managing and
 - analyzingmarketing performance to maximize its effectiveness and optimize return on investment (ROI).
- It allows to monitor campaigns and their respective outcomes, enabling companies to spend each dollar as effectively as possible.

- Beyond the obvious sales and lead generation applications, marketing analytics can offer profound insights into **customer preferences** and **trends**.
- Despite these compelling benefits, a majority of organizations fail to ever realize the promises of marketing analytics.
- According to a survey of senior marketing executives published in the Harvard Business Review,

"more than 80% of respondents were dissatisfied with their ability to measure marketing ROI."

The Need of Marketing Analytics

- A simple question with even simpler answer - **“To get the most out of your available resources”**.
- Marketing analytics helps to re-structure unorganized data into valuable insights using attribution.
- It puts the data in a visualization mode, revealing customer journeys across devices, geographies and more.

Need and importance of Marketing Analytics

- **Critical in making marketing budget**
 - Marketing is a significant expense and leaders want to know exactly what they're getting for it.
 - Marketing analytics helps in setting up a detailed roadmap that outlines the strategies, costs and projected results over a period.
 - It's also a great opportunity to focus on the future, generate new ideas, and inspire the team

- **Deciding where to spend**

- Marketers often calculate ROI at the channel or campaign level to determine **which efforts have a higher return** and therefore deserve additional investment.
- When **experimenting with a new ad network or marketing channel**, a crucial first step for marketers is **setting up analytics** such that marketing teams can measure ROI and determine if the network or channel is driving performance.
- Thus, a critical piece of the decision making, **to spend wisely and on right channels** can be driven by analytics.

- **Smart Optimization**

- Marketers often **optimize ongoing campaigns on the fly** to maximize performance based on the analytics reports.
- Effective optimizations require **digging deeper into spend and performance data**, to achieve more granular levels of reporting.
- Marketers might also seek to inform their optimizations with ROI analysis at the **Campaign, Creative, Keyword, Geographic or User level** — or any combination of the aforementioned dimensions.

3.1.1 Key Marketing Engagement & ROI metrics

Brand Awareness:

- This metric is used to show brand awareness.
 - It is evaluated based on the traffic driven to website,
 - how much people share the content,
 - the value of mentions and the reach of social messaging.
- Learn about brand awareness from **survey findings** and the amount of **news coverage over time**.
- Brands must measure it against **some kind of baseline**, which may include the historical data captured year over year.

Lead Generation

- This metric identifies **potential customers who may be interested in the products or services** being offered.
- Many companies use **dedicated landing pages on their websites** to capture information from these leads, for example: an email address, mailing address or phone numbers.
- Brands may want to deliver metrics on the average number of **leads generated weekly or monthly**.
- Some companies use **lead generation around specific events**, both online and in person, such as a **webinar or a trade show** to capture this metric.
- Lastly, brands may also **track the percentage of leads brought in** over a period versus the percentage of those **that converted to a sale**.

Customer Acquisition:

- **Bringing in new customers** is one of the best ways to show the success of marketing campaigns.
- However, every customer comes with the **cost of acquisition**.
- Add up **all the marketing expenses** and then divide this by **number of customers brought** into the business over a period.
- Other customer acquisition metrics include the **time it takes to close the sale** and the **average sale value**, which is sales profit divided by the number of sales converted.

Customer Loyalty:

- One of the other big goals is **to retain the customer**.
- This metric is used to show the **number of loyal customers** and also includes the **retention rate of the customers**.

Lifetime Value:

- It is a measure that predicts customer spending.
- How much **gross profit** can be attributed **to a particular customer** for the duration or the future of relationship.
- While measuring the campaign goals, brands must make sure that they **track metrics over time** and always **benchmark them against past performance or competitors**- if the data is available.

Digital Marketing & Transformation (BCAE0005)



Lecture -29

Anuj Mangal

anuj.mangal@gla.ac.in

+91-9897534383

3.1.2 Introduction of data science and analytics concepts

- Data science is a broad term that refers to a collection of related discipline data, focusing on the use of data to create new information and knowledge.
- The ultimate goal is to provide useful insights for better marketing decisions.
- Computers have helped to automate the routine task of number crunching and have freed us to concentrate on more creative and meaningful aspects of data science such as setting up a model and interpreting the results of a data analysis.
- However, they have also contributed to the production of significantly more data by becoming data sources themselves.

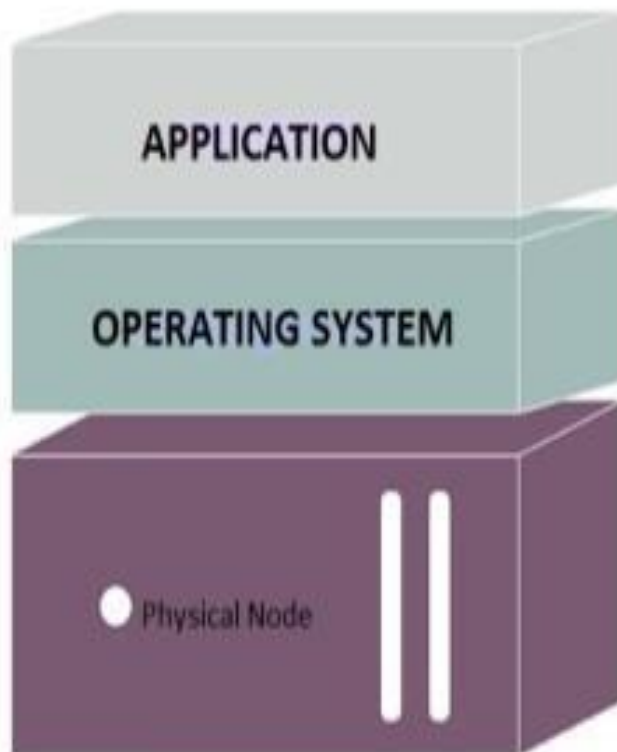
Below are some common trends to study data science:

Virtualization

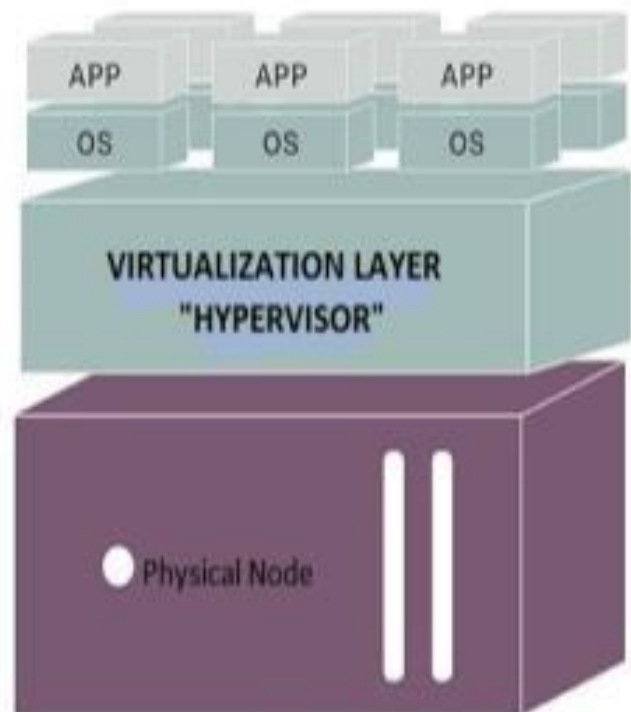
- Virtualization is the creation of a virtual -- rather than actual -- version of something, such as an operating system, a server, a storage device or network resources.
- You probably know a little about virtualization if you have ever divided your hard drive into different partitions. A partition is the logical division of a hard disk drive to create, in effect, two separate hard drives.

The evolution of virtualization

- Operating system virtualization is the use of software to allow a piece of hardware to run multiple operating system images at the same time.
- The technology got its start on mainframes decades ago, allowing administrators to avoid wasting expensive processing power.
- Virtualization platforms like VMware and Microsoft HyperV make this possible.



TRADITIONAL ARCHITECTURE



VIRTUAL ARCHITECTURE

Cloud Computing

- The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer.

Machine Learning

- **Machine learning** is an application of artificial **intelligence** (AI)
- It provides systems the ability to automatically learn and improve from experience without being explicitly programmed.
- It assists in things like sales forecasting, customer segment analysis, insurance claim fraud detection, etc.

3.2.3 Web Traffic nuances

- The amount of traffic and the **details of each visit** are extremely valuable information to a Web-based business.
- The server computer **records every request for a Web page**, allowing its operators to determine which pages get the most attention.
- Web traffic analysis gives businesses concrete, reliable information on the **interests of their customers**.

Evaluation and management of Web Traffic

Sessions:

- When someone visit a Web site, they typically browse through a few different pages before moving on to other sites.
- Web engineers call this visit a session.
- It has a well -defined beginning and an end, though the details vary from one session to the next.

Hits:

- Each Web page may consist of dozens of separate files.
- The server sends each file to your browser, assembles them into a finished piece complete with text and graphics.
- Each file sent constitutes one “hit,” so a single page view results in many hits.
- The more traffic a Web site receives, the more sessions and hits are recorded by servers.

Server Log:

- Every time a Web server processes a file request, the computer makes an entry into a server log, a dedicated file on the server's hard drive.
- The entry contains the Internet address of the user requesting the file, the time and date of the request and the file's name.
- The log may also contain error codes, such as when a user requests a non-existent page.
- The log accumulates many thousands of entries over time, becoming a gold mine of information.
- The site's operators use Web traffic analysis programs to study through the log.
- The analysis reveals the most successful pages on the site, the days and times when users are busiest and from what parts of the world the visits originate.

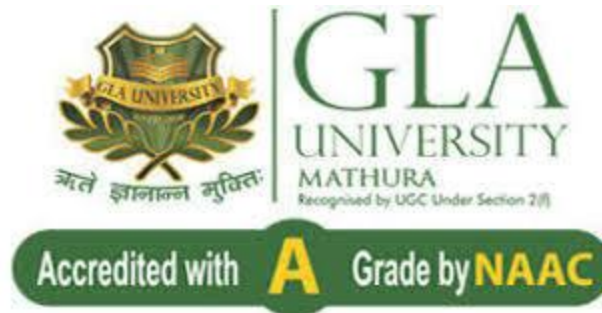
Capacity:

- Companies can host a modest site on a single server computer and an economical network connection, but a busy site may require dozens of computers and a network that can keep up with the requests.
- Web traffic requires capacity.
- A given site's operators monitor the traffic of the server's processes and use this information to determine when to add more capacity.
- If the operators neglect site capacity, floods in traffic overload the computers and slow the site's responsiveness, resulting in frustrated customers and lost business.

Traffic Analysis:

- Determines the effectiveness of business strategies.
- For example, a Web site offering a sale on certain goods can measure the traffic driven by sales.
- The analysis may reveal that the sale pricing wasn't aggressive enough, or it may show customers became interested in other products offered on the site.

Digital Marketing & Transformation (BCAE0005)



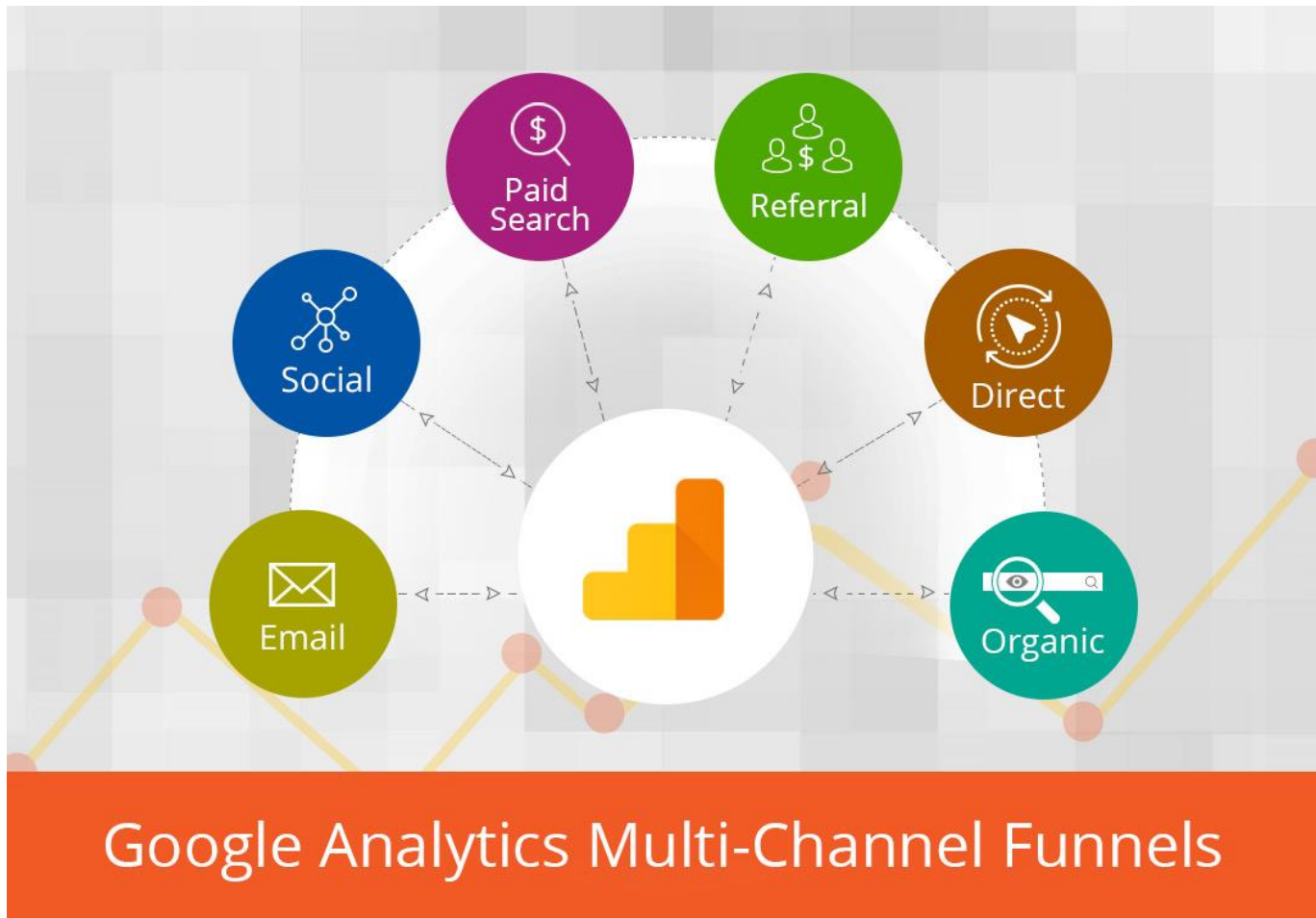
Lecture -30

Anuj Mangal

anuj.mangal@gla.ac.in

+91-9897534383

3.2.4. Multichannel Analytics



3.2.4. Multichannel Analytics

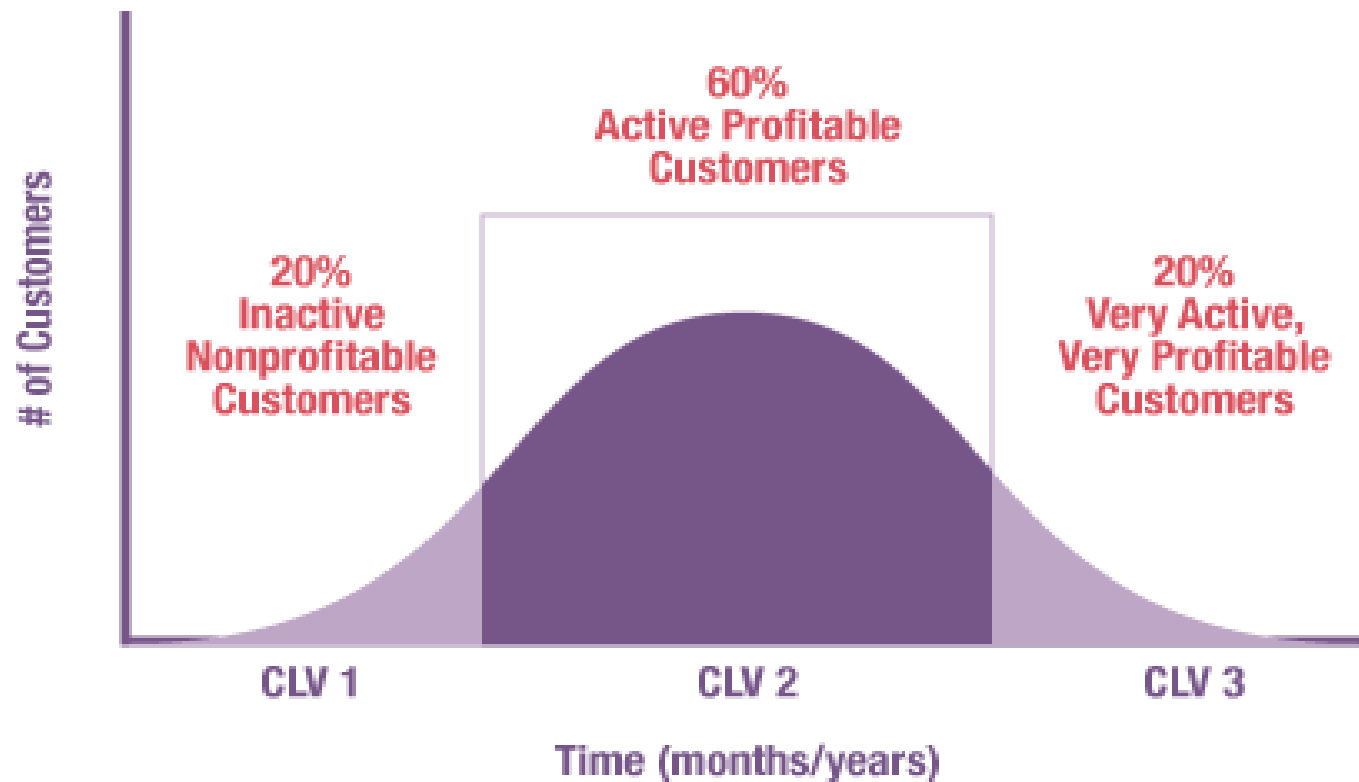
- Multichannel analytics is the practice of getting information from various marketing channels, such as
 - email,
 - traditional,
 - social,
 - offline, etc.,and then combining these pieces of information into one software environment that allows comprehensive reporting and analysis.

- **Goal of multichannel analytics:** is to enable companies to gain valuable business intelligence about their customers and prospects.
- It allows companies
 - to segment customers more efficiently
 - What content and offers are to be sent,
 - when they should be sent and
 - what should be the preferred channels.

3.2.5 Decoding Customer lifetime value CLV and RFM

- It highlights how many customers will have a **long-term relationship** with the business.
- A one-off purchase might show a high cost per acquisition, but if considering repeat purchases, that cost per acquisition creates longer term value for your business.
- It's true that it's much easier to keep an existing customer than it is to win a new one.
- Selling more to an existing customer will also be at a reduced marketing cost.

Customer Lifetime Value



Two major challenges with CLV

- The biggest and most glaring issue with CLV is the use of the word lifetime. What exactly is a lifetime?
- It's not 80 years.
- Do we really need to wait until someone has died before we can calculate the customer lifetime value of our customers? Of course not, and that's why it's important for every business to determine what the specific lifetime is.
- For a business, a customer may transact for an average of two years, so the lifetime would be two years.
- If the customer transacts on average for 15 years, 15 years would be the lifetime in this case. After specifying how long the customer lifetime is, companies have a good estimate of when the customers should be transacting, and more importantly, if they haven't.
- This information will better equip you to know the difference between a customer retained and a customer lost.

- The second challenge with CLV is there is no such thing as an average customer.
- It can be dangerous to lump your customers into this big amorphous blob in today's day and age.
- Companies can use averages but it's important to understand that every customer is a person and every person deserves to be treated a little bit better than just an average.

Calculation of CLV

- Most efficient and the easiest model to calculate CLV would be to:
 - take the total profit of a client over the lifetime of buying and subtract all the sales, marketing, and service delivery cost from that client.
- The number you have left is your customer lifetime value number.

Example of CLV calculation

- For example, suppose a new customer brings you **100 dollars in profit on their first purchase.**
- That same customer on average **purchases two more times from you over the next two years.**
- Each time they reorder, they spend approximately 250 dollars with an addition of **100 dollars in profit on each purchase.**
- Total profit per customer during CL = ???
- Therefore, if we do a simple math, that **customer is worth 500 dollars in profit per their lifetime.**

- Let's suppose you **run pay-per-click Google ads** and it **cost you approximately 75 dollars in clicks** to generate that first sale.
- **Net profit = ????**
- Now you might think because the customers spent 100 dollars and it costs you 75 dollars, you only generated 25 dollars in profit, but since you know the customer lifetime value of your customer's 500 dollars, you know that 75 dollars just generated an additional **425 dollars for your business**.

Recency, Frequency and Monetary RFM Calculation Model

- To identify customer lifetime value, a well-known method called recency, frequency and monetary (RFM) model is used to **represent customer behavior characteristics**.
- RFM models have been **used in direct marketing for more than 30 years**.
- These models were developed to target marketing programs at specific customers with the **objective to improve response rates**.
- Prior to these models, companies **typically used demographic profiles of customers** for targeting purposes.
- However, research strongly suggests that past purchases of consumers are better predictors of their future purchase behavior than demographics.

- The basic assumption of using the RFM model is that **future patterns of consumer trading resemble past and current patterns.**
- This study proposes using the following RFM variables
 - **R (Recency)** : the period since the last purchase;
(a lower value corresponds to a higher probability of the customer's making a repeat purchase;)
 - **F (Frequency)**: number of purchases made within a certain period;
(higher frequency indicates greater loyalty;)
 - **M (Monetary)**: the money spent during a certain period;
(a higher value indicates that the company should focus more on that customer)

3.2.6 Deciphering Conversion and Goal Metrics

- Conversion tracking must be present, since that will enable to assign a value to the campaign results.
- It is as easy as adding a piece of PHP code to the website, so the developer can do this in no time.
- Many websites have a standard conversion code installed on all the pages, and with Analytics companies specify which pages to track and count as a conversion.
- Below are some pages which could count as conversion:
 - **a thank-you page**
 - **a receipt page**
 - **a submission-confirm page**
 - **a registration-complete page**

- With Google Analytics, companies can designate specific pages to be tracked,
 - The amount of time a user spent on a site and
 - The number of page views and other various other onsite behaviors.
- They can specify all these metrics as a goal, and then track these goals accurately over time.

How Google Analytics Work

- By understanding how Google Analytics works, companies will be better equipped to identify problems with the setup or understand why certain data anomalies might occur.
- There are four components to how the Google Analytics system works:
 - Data Collection
 - Configuration
 - Data Processing
 - Reporting.

- To track a website, Google Analytics uses a small snippet of **JavaScript code**.
- Developers **place this code on every page of the website** and when a user arrives, the code comes alive and starts **to collect data on how the user engages with the site**.
- Some of the **data comes from the website itself, like the URL of the pages** that the user is viewing.
- Other **data is collected from the user's browser**, like the language the browser is set to, the browser name, and the device and operating system used to access the site.
- The JavaScript can also collect information on **what content is viewed, for how long, and even the referring source** that brought the user to the site.

3.2.7 Implement conjoint analysis & decision tree tactics

- Conjoint analysis involves the measurement of psychological judgments (such as consumer preferences, or acceptance) or perceived similarities or differences between choice alternatives,
- It provides the marketers a greater insight into future predictions and make way through the decision tree.
- The name "Conjoint analysis" implies the study of the joint effects.
- In marketing applications, we study the joint effects of multiple product attributes on product choice.
- Marketing managers are faced with numerous difficult tasks directed at assessing future profitability, sales, and market share for new product entries or modifications of existing products or marketing strategies.

CONJOINT ANALYSIS

Which of these laptops would you choose?

Brand	HP	Dell	Sony
Processor	4GHz	3GHz	2GHz
RAM	1GB	3GB	2GB
Monitor	21-inch	17-inch	15-inch
Price	\$899	\$799	\$699
	☐	☐	☐

- These specific tasks include predicting:
 - The **profitability and/or market share for proposed new product** concepts given the current offering of competitors
 - The **impacts of new competitor products on profits** or market share if we make no change in our competitive position
 - Customer switch rates either from our current products to new products we offer (cannibalism), or from our competitors' products to our new products (draw)
 - The differential response of above items by key market segments purchasing our product
 - Competitive reaction to our strategies of introducing a new product. Specifically, should a new product be introduced, and if so, what is the optimal design configuration for this new product?

3.2.8 Avoiding Common Analytical Pitfalls

- The most common pitfall is jumping to answers too quickly.
- Some of the tools they can use to avoid these pitfalls are - defining your problem well & having a clear problem statement.

- The second pitfall is being unwilling to expand the problem space.
- Sometimes they're going to focus on things that don't matter.
- They'll come up with a recommendation, and even though it's insightful, it won't really change the performance of the organization

- By leveraging the technique of the 80/20 Principle, you can ask yourself, "If I solve this, will it have a unbalanced impact on my organization?" "Or, am I solving for that 80% that drives 20% of the results that don't really matter?"